

CS 3110 HW1

1) Prove or disprove the following statements

a) $(3n^2 + 1)(4n^3 + n)/2 \in O(n^5)$

Let us assume that $f(n) = (3n^2 + 1)(4n^3 + n)/2$ and $g(n) = n^5$

$$(12n^5 + 3n^3 + 4n^3 + n)/2$$

$$= (12n^5 + 7n^3 + n)/2$$

$$= (12n^5 + 7n^5 + n^5)/2$$

$$= (20n^5)/2$$

$$= 10n^5$$

$$\leq c \cdot g(n)$$

So let $c = 24$ and $n_0 = 1$, then $\forall n \geq 1, f(n) \leq c \cdot n^5$, so $f(n) \in O(g(n))$.

Hence Proved

b) $2^{2n+2} \in O(2^{2n+1})$

Let us assume that $f(n) = 2^{2n+2}$ and $g(n) = 2^{2n+1}$

$$2^{2n+2} = 2 \cdot 2^{2n+1}$$

$$\leq c \cdot 2^{2n+1}$$

So let $c = 2$ and $n_0 = 1$, then $\forall n \geq 1, f(n) \leq c \cdot 2^{2n+1}$, so $f(n) \in O(g(n))$

Hence proved

c) For $a > 1$ and $b > 1$

$$O(\log_a n) = O(\log_b n)$$

$$\log_a n = \log_b n / \log_b a$$

$$= 1/\log_b a \cdot \log_b n$$

$$= x \cdot \log_b n \text{ (where } x \text{ is } 1/\log_b a)$$

Now let us prove that $O(\log_a n) \in O(\log_b n)$

Let us assume that $c = \max(x, 1)$ (we are using max so that we can take any one of the larger values amongst x and 1 and assign it to c to do the question) and n_0 as 2 (since $\log 1 = 0$), so we can say that for $\forall n \geq n_0$:

$$\log_a n = x \cdot \log_b n \leq c \cdot \log_b n$$

Hence, we can say that $\log_a n \in O(\log_b n)$
 Now, let us prove that $O(\log_b n) \in O(\log_a n)$

$$\begin{aligned}\log_b n &= \log_a n / \log_a b \\ &= 1/\log_a b * \log_a n \\ &= C * \log_a n, \text{ where } C = 1/\log_a b\end{aligned}$$

Let us assume that $a = \max(C, 1)$ and n_0 as 2 (since $\log 1 = 0$), so we can say that for $\forall n \geq n_0$:

$$\log_b n = C * \log_a n \leq a * \log_a n$$

Hence, we can say that $\log_b n \in O(\log_a n)$

Since we were able to prove that $\log_a n \in O(\log_b n)$ and $\log_b n \in O(\log_a n)$, so we can say that $O(\log_a n) = O(\log_b n)$.

2) Derive the runtime of the following code as a function of n and determine its Big-O upper bound. You must show the derivation of the end result.

a) Inner Loop

Let us assume that the inner loop takes c_1 time for setup and c_2 for doing the runtime and performing all operations inside. So, we can say that:

$$\begin{aligned}T_{\text{inner}} &= c_1 + \sum_{k=1}^{i+j} c_2 \\ &= c_1 + c_2(i+j)\end{aligned}$$

Middle Loop

Let us assume that the inner loop takes c_3 time for setup and c_4 for doing the runtime and performing all operations inside. So, we can say that:

$$\begin{aligned}&= c_3 + \sum_{j=1}^i (c_4 + T_{\text{inner}}) \\ &= c_3 + \sum_{j=1}^i c_4 + \sum_{j=1}^i T_{\text{inner}} \\ &= c_3 + c_4 i + \sum_{j=1}^i (c_1 + c_2(i+j)) \\ &= c_3 + c_4 i + \sum_{j=1}^i c_1 + \sum_{j=1}^i c_2 i + \sum_{j=1}^i c_2 j \\ &= c_3 + c_4 i + c_1 i + c_2 i^2 + c_2(i*(i+1))/2 \\ &= c_3 + (c_4 + c_1)i + c_2 i^2 + (c_2 i^2 + c_2 i)/2 \\ &= (2c_3 + (2c_4 + 2c_1 + c_2)i + c_2 i^2)/2\end{aligned}$$

Outer Loop

Let us assume that the inner loop takes c_5 time for setup and c_6 for doing the runtime and performing all operations inside. So, we can say that:

$$\begin{aligned} & c_5 + \sum_{i=1}^n c_6 + \text{middle loop} \\ &= c_5 + \sum_{i=1}^n c_6 + \sum_{i=1}^n \text{middle loop} \\ &= c_5 + \sum_{i=1}^n c_6 + \frac{1}{2}(\sum_{i=1}^n 2c_3 + (2c_4 + 2c_1 + c_2)i + 3c_2i^2) \\ &= c_5 + c_6n + \frac{1}{2} \sum_{i=1}^n 2c_3 + \frac{1}{2} \sum_{i=1}^n (2c_4 + 2c_1 + c_2)i + \frac{1}{2} \sum_{i=1}^n 3c_2i^2 \\ &= c_5 + c_6n + c_3n + \frac{1}{2} (2c_4 + 2c_1 + c_2)(n(n+1)/2) + \frac{3}{2} (n(2n+1)(n+1)/6) \\ &= c_5 + (c_6 + c_3)n + ((2c_4 + 2c_1 + c_2)(n^2 + n))/4 + (2n^2 + n)(n + 1)/4 \\ &= c_5 + (c_6 + c_3)n + ((2c_4 + 2c_1 + c_2)(n^2 + n))/4 + (2n^3 + 2n^2 + n^2 + n)/4 \\ &= (4c_5 + (2c_4 + 2c_1 + c_2 + 3)n^2 + (2c_4 + 2c_1 + c_2 + 1)n + 2n^3)/4 \\ &= O(n^3) \end{aligned}$$

b) Inner Loop

Since the inner loop is going to run 300 runtimes, we can say that its runtime will be $O(1)$.

Outer Loop

Let us assume that each iteration of the loop does c_1 operations, so after k iterations we can say that

After iteration	Value of i
1	5
2	$25 = 5^2$
3	$125 = 5^3$
....	
k	5^k

Since the loop ends when i becomes equal to n , we can say that:

$$5^k = n$$

$$k = \log_5 n \text{ (if } a^k = n, \text{ then } k = \log_a n)$$

Since we know that the runtime for the inner loop is $O(1)$, $k = \log_5 n$ iterations and each iteration does c_1 operations, we can say that the total runtime can be:

$$c_1 * (\log_5 n * 1) \in O(\log_5 n)$$

c) Power function

According to the question, the runtime of the power function is $O(n)$.

Outer Loop

Let us first determine the no. of iterations of the while loop, so that we can take out its runtime. Let us assume that the while loop has k iterations.

After iteration	Value of i
1	2^n
2	2^{n-1}
3	2^{n-2}
....	
k	2^{n-k}

Since we know that the loop runs until the value of i is greater than or equal to 1, we can determine k in the following way:

$$\begin{aligned} 2^{n-k} &\geq 1 \\ n - k &\geq \log 1 \\ n - k &\geq 0 \\ n &\geq k \end{aligned}$$

Therefore we can say that the while performs $n + 1$ iterations, since the loop terminates after k iterations, hence the runtime the while loop is $O(n+1) = O(n)$

Inner loop

Since the value changes after each iteration of the for loop, its runtime is going to be different every time, so we can calculate it in the following way:

$$2 * \sum_{k=0}^n 2^{n-k}$$

Let us assume an integer m such that $m = n - k$, so that means when $k=0$, $m=n$ and when $k=n$, $m=0$, so if we reverse the limits we can say that:

$$2 * \sum_{m=0}^n 2^m = 2 * (2^{m+1} - 1) = 2^{m+2} - 2$$

Hence, we can say that the runtime of the inner loop is $O(2^n)$ and since it is greater than the runtime of the while loop [$O(n)$], we can say that the worst case runtime will be the runtime of the for loop. Hence, the total runtime is:

$$O(2^n) + O(n) \text{ (runtime of the power function)} = O(2^n)$$